scores of the Samsung ATA 133 drives. We need to remind you that the transfer speeds also depend on the test bed hard drive, which happens to be faster than the two Samsung ATA 133 drives. Hence the read speed was more than the write speed for these two drives. We would recommend you rely on the intra-drive speed to measure the performance of these drives. Exceptional performers in the inter-drive test were the Hitachi Deskstar 250 GB and the WD Caviar RE2.

As expected, Samsung's parallel ATA drives were beaten by a big margin in the intra-drive transfer test. The Samsung SP0411C, a SATA drive with a 2 MB cache, performed no better than the Samsung ATA drives! The WD Caviar RE2, Hitachi Deskstar 500 GB and Maxtor's Maxline III topped the charts with superior times in the intra-drive test.

The Photoshop image load test is one of the most important tests, especially for those using software such as Maya, 3D Studio and Photoshop. Photoshop's scratch disk, or the virtual RAM space for the application, was created on the test sample. Images of varying sizes (200 MB, 550 MB, 800 MB, 1 GB and 1.4 GB) were also stored on the test sample, so as to maintain the outcome of the test purely on the basis of the performance of the test sample. We used Photoshop to open these image files from the test hard disk and timed the process for each image.

The contest for the top slot was fierce, between the WD Caviar SE (WD2500KS) and the Hitachi Deskstar 500 GB, with the latter recording better timings. Other drives that performed well were the Maxtor MaxLine III and Hitachi Deskstar 250 GB. If you're looking for a performer at an affordable budget, then the WD2500KS is your best bet!

Value For Money

Rs 2,500 and Rs 4,000 are attractive figures, but have you thought about how much you're spending per GB of storage? It's a fair notion if you're a moderate user, but the actual price is not an ideal criterion if you demand performance at a fair price.

Back to the paperwork: all the drives have a remarkable warranty period—either three or five years—so the main aspect of discussion is the price index. Let's start with the Samsung hard drives, which we introduce again and again as the reference. The SP0411 and SV0411N belong to the category that most people buy. The price that one pays per GB for this category of hard disk is the highest—you can get double the capacity (80 GB) and much better performance for just an additional 1,000 rupees!

Maxtor's DiamondMax10, with a capacity of 300 GB and priced at Rs 8,000, proved to be the most cost-effective of all the drives, at Rs 28.62 per GB. On the other hand, WD's 74 GB Raptor priced at Rs 8,200 is the most expensive, at a whopping Rs 118.42 per GB.

Generally, we see that the cost per GB of higher capacity drives is much less than that of lower capacity ones, but the Seagate Barracuda (400GB) ripped through the price paradigm. It was not only costlier than its counterparts, the WD4000KD and WD4000YR, but was also more expensive than the Hitachi Deskstar 500 GB by a margin of Rs 3,500.

Considering the warranty, price index and performance, the Hitachi Deskstar 80 GB, Maxtor MaxLine III 300 GB, Western Digital WD2500KS 250 GB and Hitachi Deskstar 250 GB are the ones that make for value for money, in their respective categories. However, if you are hunting for a 160 GB drive, the Samsung HD160JJ is your best bet.

The Consensus

Based on our results from the Performance comparison and the overall analysis, we award the Best Buy Gold to the test sample that secured the highest overall scores.

The overall score is obtained by giving appropriate weightages to individual test segment scores. A higher weightage was given to the intra-drive test and the application load time test in the Performance segment, while additional features and shock resistance were the determinants in the Features segment. Hitachi's Deskstar 500GB recorded the highest performance scores amongst all the drives but it lost out on price index.

Western Digital Caviar RE2 WD4000YR-01PLBO

We have two Best Buy Gold winners, the Hitachi Deskstar 250 GB (HDT722525DLA380) and the WD 250 GB (WD2500KS). It was difficult to judge the better of the two as warranty, price and capacity were identical. While the Hitachi leads in the performance segment, the WD2500KS leads in features.

The WD Caviar RE2 was close to Gold with better performance and features than the two Gold winners, but had to settle for Best Buy Silver due to its slightly higher cost per GB.

What Next?

Along with increased capacities, there has been a remarkable decrease in the dimensions of hard disks. The storage density of the platters has increased as well. There had to be a limit to the prevailing technology, and researchers had to think of new ways to develop a high-volume storage medium.

Thus, the need for higher capacity and ace performance is changing the way data is written onto a medium. Currently, data is written onto the disk surface by Longitudinal Recording (LR), which is reaching its limits (refer *Towards Terabytes*, *Digit*, October 2005). Disk manufacturers are now moving on to Perpendicular Recording, which promises much higher densities.

We are anxious to get our hands on this technology, and we're sure you will be looking forward to it as well! ⊠

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